

Solution Requirements

Date	29 September 2026
Team ID	6ab4fddea238dc999eb665e9
Project Name	PocketSmart AI: Your Smart Budget & Recommendation Assistant
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should do).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should be).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users shall register and securely log in using password hashing and JWT-based authentication/session handling. Logout shall end the active session.	High
2	Authorization levels	Protected planning, recommendation, details and history functions require an authenticated user. Separate administrator roles are not required for the current prototype.	Medium
3	External interfaces	Integrate the configured Gemini AI service for recommendations and provide safe search/demo links for relevant platforms such as Amazon, Flipkart, IKEA, Swiggy, Zomato and OYO. API keys remain in environment variables.	High

	Requirement Category	Requirement Description	Priority (High/Medium/Low)
4	Transactions processing	Collect budget, preferences and planning inputs; validate the requested budget; generate recommendations; and store recommendation/session history. Real payment processing is not required.	High
5	Reporting	Provide dashboard and history views so users can review previous recommendations and open recommendation details, categories and estimated costs.	Medium
6	Business rules, etc.	Support Home Interior, Party Planning and Jewelry Planning. Recommendations must respect the stated budget. Jewelry planning may accept an outfit image for multimodal analysis when configured.	High
7	Compliance to laws or regulations	Protect credentials and API secrets, validate uploaded files, and use user-provided data only for the application's planning and recommendation functions.	High
8	Other	Provide safe fallback/mock recommendation data when Gemini fails, validate structured AI output, and show clear user-facing errors.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
------	--------------	-------------------------	-------------------------------------

	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Normal pages should respond promptly; AI response time depends on the configured Gemini service.	Target: normal pages load AI latency is service-depen
2	Scalability	Separate authentication, database, schemas, AI utilities and routes so additional planning domains can be added without redesigning the application.	Acceptance: new planners added modularly.
3	Security & Data Privacy	Hash passwords, keep JWT/session secrets and Gemini keys in environment variables, validate uploads, and never commit secrets to Git.	Acceptance: no plaintext p or hard-coded API secrets uploads are rejected.
4	Reliability & Availability	Core planning flows should remain usable when Gemini is unavailable through the implemented safe fallback path, with clear service limitations.	Acceptance: Gemini failure crash the application; fallback response is returned.
5	Usability & Accessibility	Provide clear separate planners for Home Interior, Party and Jewelry, with simple budget inputs, recommendations, details and history navigation.	Acceptance: a user can co planner flow without knowi implementation.
6	Other	Maintain a documented FastAPI/Jinja2 structure with requirements, environment template, README and automated tests.	Acceptance: setup and tes instructions are documente project runs from the supp